

Multiple Choice (10 marks)

1. If $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}$, for all $a, b \in G$, then G is a/an
 2. commutative semi group
 3. abelian group
 4. non-abelian group
 5. None of these

If $A = \{(1, 2), (1, 3), (4, 2)\}$. Let $\sim = \{(1, 2), (1, 3), (4, 2)\}$. Then $\sim$ is

- (a) not anti-symmetric
- (b) transitive
- (c) reflexive
- (d) symmetric

Find all zeros in the indicated finite field of the given polynomial with coefficients in that field. $x^5 + 3x^3 + x^2 + 2x$ in $\mathbb{Z}_5$

- (a) 0
- (b) 1
- (c) 0,1
- (d) 0,4

A group G in which $(ab)^2 = a^2 b^2$ for all a, b in G is necessarily

- (a) finite
- (b) cyclic
- (c) of order two
- (d) abelian

Find the maximum possible order for an element of S_n for $n = 7$.

- (a) 6
- (b) 12
- (c) 30
- (d) 105

True / False (10 marks)

Answer True or False

6. True or False: If $f(1 + x) = f(x)$ for all real x and $f(5) = 11$, then $f(15/2)$ is also equal to 11.
7. True or False: Every cycle is a permutation of a set of elements.
8. True or False: The circle defined by $x^2 + y^2 = 16$ intersects the parabola $x^2 = y + 4$ at the greatest number of points.
9. True or False: The product $(2,3)(3,5)$ in the ring $\mathbb{Z}_5 \times \mathbb{Z}_9$ is equal to $(1,6)$.
10. True or False: Every permutation is a one-to-one function from a set to itself.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of cardinality in set theory and analyze why the set of all functions from $\mathbb{R}$ to $\{0, 1\}$ has the greatest cardinality among various sets.
12. Discuss the properties of the trace function for $n \times n$ matrices. Analyze the validity of the statements regarding the trace of A^2 , the trace of idempotent matrices, and the trace of matrix products.

End of Paper